

BRAINS COLLEGE BAGHBANPURA CAMPUS

PYTHON 12 WEEK COURSE OUTLINE

WEEK 1: INTRODUCTION | INSTALLATION OF PYTHON

Lecture 1	Introduction, Installation and Setup, Write your first program in python
Lecture 2	Basic Syntax and Data types in python (int,float,complex,bool,str), take user input
Lecture 3	Variable, Operator (arithmetic, relational, assignment, logical), Reserved Words, Comment
Lecture 4	Type Casting, Type Conversion, Madlibs game
Lecture 5	Q1. Write a Program to input 2 floating point numbers & print their average. Q2. Write a Program to input 2 numbers & print their sum. Q3. Calculator in python part -1
Lecture 6	Q4. Calculator in python part -2

WEEK 2: PROGRAMMING IN PYTHON (1)

Lecture 7	Conditional Statements. Q1. Write a Program to check if a number entered by the user is odd or even.
Lecture 8	Strings, String Functions, Indexing Q1. Write a Program to input user's first name & print its length. Q2. Write a Program to find the occurrence of '\$' in a String
Lecture 9	Lists, Lists Functions, List Slicing Q1. Write a Program to ask the user to enter names of their 3 favorite movies & store them in a list.
Lecture 10	Tuples, Tuple Functions Q1. Write a program to count the number of students with grade "A" in following tuple. ["C","D","A","A","B","B","A","A"]
Lecture 11	Dictionary, Dictionary Functions Q1. Write a Program to enter marks of 3 subjects from the user and store them in a dictionary. Start with an empty dictionary & add one by one. Use subject name as key & marks as value
Lecture 12	Sets, Set Functions Q1. Figure out a way to store 9 & 9.0 as separate values in the set. (You can take help of built-in data types)

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WEEK 3: PROGRAMMING IN PYTHON (2)

Lecture 13	While loop, break and continue statements. Q1. Print all numbers from 1 to 100 and 100 to 1.
Lecture 14	For loop, for else loop Q1. Search for a number x in the this tuple using loop [1,4,9,16,25,36,49,64,81,100]
Lecture 15	range(), pass statement Q1. Print the multiplication table of 7 using for() and range().
Lecture 16	Functions, built-in and user defined functions, default parameters Q1. Write a function to convert USD to PKR
Lecture 17	Recursive functions in python Q1. Write a recursive function to calculate the sum of first n natural numbers
Lecture 18	Make a countdown timer program in python

WEEK 4: ADVANCED PROGRAMMING IN PYTHON (1)

Lecture 19	File I/O part-1
Lecture 20	File I/O part-2
Lecture 21	Membership Operators, modules
Lecture 22	Object Oriented Programming part-1
Lecture 23	Object Oriented Programming part-2
Lecture 24	Q1. Make an alarm clock

WEEK 5: ADVANCED PROGRAMMING IN PYTHON (2)

Lecture 25	Exception Handling (try,catch,finally)
Lecture 26	Python packages and pip installation
Lecture 27	Working with Date and Time module
Lecture 28	Number Guessing Game
Lecture 29	Create a marksheet generator
Lecture 30	Revision

WEEK 6: DATABASE AND GUI DEVELOPMENT

Lecture 31	Introduction to Databases
Lecture 32	SQLite Database in Python
Lecture 33	CRUD Operations (Create, Read, Update, Delete)
Lecture 34	GUI Development using Tkinter
Lecture 35	Q1. Make a calculator GUI
Lecture 36	Revision

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WEEK 7: INTRODUCTION TO AI & DATA SCIENCE

DAY	WORK
Lecture 37	What is AI, Machine Learning, Deep Learning
Lecture 38	Introduction to Data Science
Lecture 39	Introduction to NumPy
Lecture 40	NumPy Arrays and Operations
Lecture 41	Introduction to Pandas
Lecture 42	Working with CSV files

WEEK 8: DATA VISUALIZATION + AI BASICS

Lecture 43	Data Cleaning using Pandas
Lecture 44	Introduction to Matplotlib
Lecture 45	Bar Charts, Pie Charts, Line Graphs
Lecture 46	AI Concepts: Training Data, Prediction, Model
Lecture 47	Introduction to Scikit-learn
Lecture 48	Salary Prediction program

WEEK 9: BASICS OF MACHINE LEARNING

Lecture 49	Supervised vs Unsupervised Learning
Lecture 50	Linear Regression
Lecture 51	Classification Basics
Lecture 52	Train/Test Split
Lecture 53	Accuracy Checking
Lecture 54	House price prediction program

WEEK 10: COMPUTER VISION + AI APPLICATIONS

Lecture 55	Introduction to OpenCV
Lecture 56	Reading Images in Python
Lecture 57	Face Detection Basics
Lecture 58	Image Filters and Processing
Lecture 59	AI Applications in Real Life
Lecture 60	Face Detector App

WEEK 11: CHATBOT DEVELOPMENT

Lecture 61	What is a Chatbot?
Lecture 62	Rule-Based Chatbot using Python
Lecture 63	Introduction to APIs
Lecture 64	Using Gemini API or OpenAI API
Lecture 65	Building AI Chatbot
Lecture 66	Testing and Debugging

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WEEK 12: FINAL PROJECT + DEPLOYMENT

Lecture 67	Chatbot Project Completion
Lecture 68	GUI/Web Interface for Chatbot
Lecture 69	Introduction to Streamlit
Lecture 70	Deploying AI App
Lecture 71	Final Project Presentation
Lecture 72	revision

FINAL CHATBOT PROJECT IDEAS

Students can choose ONE:

1. College Information Chatbot
2. Fitness Chatbot
3. Python Learning Assistant
4. Motivational AI Chatbot
5. AI Career Guide Chatbot